

SKELLEFTEA, Sweden

WMO: 022930

Lat: 64.624N

Lon: 21.072E

Elev: 157

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-7.2	-18.6	2.0	-13.6	-11.1	3.0	-6.8	22.2	35.1	18.9	30.9	3.6	260	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	79.3	62.6	75.6	61.2	71.9	59.9	65.7	74.0	63.7	71.8	61.8	69.1	8.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	86.0	68.7	60.7	79.7	66.6	58.9	74.6	64.8	30.6	74.0	29.0	71.9	27.7	69.0	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.6	14.4	DB	-20.0	83.1	7.9	3.8	-25.7	85.8	-30.3	88.1	-34.8	90.2	-40.5	93.0	(4)
(5)			WB	-20.3	68.2	7.9	2.7	-26.0	70.1	-30.6	71.7	-35.0	73.2	-40.7	75.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	17.5	18.0	25.4	35.8	46.7	55.4	61.3	58.4	50.0	38.3	29.3	23.9		
	DBStd	17.31	11.78	12.91	10.00	6.29	7.56	6.38	5.46	5.62	5.95	7.06	8.77	10.30		
	HDD50	5139	1006	896	762	426	155	20	0	4	73	366	621	810		
	HDD65	9737	1471	1316	1227	875	569	295	142	217	451	828	1071	1275		
	CDD50	923	0	0	0	1	51	182	351	263	73	2	0	0		
	CDD65	47	0	0	0	0	0	8	28	11	0	0	0	0		
	CDH74	502	0	0	0	0	24	102	275	100	1	0	0	0		
	CDH80	74	0	0	0	0	1	18	43	12	0	0	0	0		
	WSAvg	6.7	6.2	6.6	7.1	6.7	7.1	7.2	6.3	6.1	6.6	6.7	6.5	6.8		
	PrecAvg	23.5	1.8	1.3	1.3	1.3	1.5	2.0	2.3	2.8	2.4	2.3	2.4	1.9		
	PrecMax	33.1	3.7	2.4	2.5	3.2	3.8	3.7	5.7	5.1	4.8	6.5	4.6	4.1		
(15) Precipitation	PrecMin	17.8	0.2	0.2	0.4	0.3	0.5	0.6	0.2	0.6	0.4	0.4	0.7	0.2		
	PrecStd	3.9	0.8	0.7	0.6	0.7	0.9	0.8	1.2	1.3	1.1	1.3	1.1	0.9		
	DB	41.7	44.5	50.1	62.7	77.3	82.6	84.1	82.2	69.9	56.8	49.6	44.5			
	MCWB	38.0	39.5	41.5	49.9	58.0	61.9	65.6	66.7	58.4	50.9	46.3	40.1			
	2% DB	37.5	41.0	45.5	55.7	71.4	76.9	80.5	77.0	65.9	53.5	46.1	41.3			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	35.1	36.7	38.8	44.8	55.6	59.1	64.2	62.0	57.1	48.6	44.0	38.3			
	5% DB	35.5	37.1	42.5	51.4	65.5	71.4	76.9	72.8	62.5	50.4	42.8	39.0			
	MCWB	34.1	34.0	36.8	42.0	52.5	57.6	62.2	60.9	55.0	47.1	40.7	37.1			
	10% DB	33.5	33.9	39.1	47.6	60.9	67.6	73.1	69.0	59.6	48.4	40.6	36.6			
	MCWB	32.6	32.7	34.5	39.6	50.2	55.3	61.6	60.2	53.9	45.7	39.1	35.1			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	38.3	39.9	41.6	50.3	59.7	64.7	69.6	68.3	60.6	52.6	46.2	40.5			
	MCDB	40.7	44.0	48.6	61.2	72.2	76.8	77.6	77.8	65.3	55.4	47.9	42.1			
	2% WB	35.3	37.0	39.2	45.3	56.6	61.4	67.1	65.0	58.5	49.9	43.7	38.7			
	MCDB	36.4	40.1	44.9	54.1	68.9	71.7	74.4	72.4	63.5	51.9	45.1	40.5			
	5% WB	33.8	34.5	37.0	42.5	53.5	59.2	64.7	63.0	56.7	47.8	40.8	36.8			
(31) Mean Daily Temperature Range	MCDB	34.7	36.1	41.4	49.9	64.2	67.6	73.1	69.7	60.8	50.0	41.8	38.3			
	10% WB	32.3	32.6	34.8	40.1	50.8	57.0	62.9	61.1	55.2	45.8	38.7	34.9			
	MCDB	33.0	33.7	37.9	46.1	59.3	64.8	70.9	67.0	58.6	48.0	39.6	36.1			
	5% DB	13.1	14.2	16.3	16.4	17.6	16.4	16.2	15.8	14.5	10.8	9.4	11.3			
	MCDBR	11.9	13.4	17.6	24.2	26.6	23.1	22.9	22.2	18.0	13.3	9.1	10.6			
(36) Clear-Sky Solar Irradiance	MCWBR	10.8	10.9	12.2	14.6	13.4	10.7	9.7	10.2	10.7	9.6	8.0	9.6			
	5% WB	11.2	11.5	16.0	22.2	25.1	19.4	19.4	17.6	15.0	11.7	8.6	10.1			
	MCDBR	10.5	9.9	11.4	13.9	13.1	10.2	9.7	9.6	10.0	9.3	8.0	9.4			
	taub	N/A	0.276	0.304	0.337	0.343	0.345	0.361	0.349	0.320	0.302	0.339	0.348			
	taud	N/A	2.166	2.233	2.308	2.459	2.496	2.473	2.500	2.558	2.401	2.607	2.633			
(42) All-Sky Solar Radiation	Ebn at Noon	N/A	197	246	265	275	277	268	260	246	192	98	47			
	Edh at Noon	N/A	18	27	31	30	30	30	26	20	15	7	4			
	RadAvg	37	215	682	1202	1570	1744	1605	1210	706	272	61	13			
(45)	RadStd	3	38	70	106	158	139	150	111	91	49	11	1			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page